Vinegar

Frank Emde, Heinrich Frings GmbH & Co KG, Bonn, Germany

ULLMANN'S ENCYCLOPEDIA OF INDUSTRIAL CHEMISTRY

1.	**Introduction**	**989**
2.	**Historical Outline**	**989**
3.	**General Aspects of Acetic Acid Fermentation**	**990**
3.1.	**Stoichiometry**	**990**
3.2.	**Microbiology**	**991**
3.3.	**Mixing and Aeration**	**993**
3.4.	**Fermentation Kinetics**	**993**
4.	**Resources of Raw Materials**	**995**
5.	**Production**	**996**
5.1.	**Processes**	**996**
5.1.1.	Submerged Fermentation	996
5.1.2.	Surface Fermentation and Trickle Processes	997
5.1.3.	Solid-State Fermentation	998
5.1.4.	Alternative Technologies and Trends	998
5.2.	**Facilities, Apparatus, and Key Equipment for Submerged Fermentation**	**999**
5.2.1.	General Plant Design	999
5.2.2.	Fermentation Systems	1000
5.2.3.	Utilities and Equipment for Up- and Downstream Processing	1004
6.	**Uses**	**1005**
7.	**Nutritional Aspects**	**1005**
8.	**Analysis**	**1005**
8.1.	**Process Analysis**	**1005**
8.2.	**Authenticity Analysis**	**1007**
9.	**Storage and Transportation**	**1007**
10.	**Legal Aspects**	**1007**
11.	**Economic Aspects**	**1008**
12.	**Environmental Impact**	**1008**
	References	**1009**

1. Introduction

Vinegar is an aqueous liquid containing significant amounts of acetic acid, which is produced in a natural fermentation process from alcohol-containing wines or mashes. Thermodynamically, the main components of vinegar — water, acetic acid, and minor amounts of ethanol — form a ternary mixture. Other substances that can be found in vinegars originate from the agricultural source, added processing aids, and blending materials or are formed during fermentation and maturation. Physical properties of raw vinegars such as density, boiling point, and viscosity do not differ much from those of water. The pH value of vinegar is between 1.7 and 4 depending on the acetic acid concentration and the buffering capacity of the other ingredients. Due to the low pH value of vinegar, microbial growth is effectively prevented.

2. Historical Outline

The production of vinegar is one of the oldest methods of foods processing. The spontaneous conversion of alcohol (in wine) to acetic acid was already used by the cultures of ancient Egypt, Babylon, India, Greece, and Rome to produce vinegar for beverage and food consumption, general household use, and medical purposes. The special hygienic properties of vinegar were particularly useful for preserving foods. As a refreshing and long-lasting drink, vinegar was very important for the hunt, war campaigns, and for journeys by boat. Medical applications of vinegar for snake bite, fever, intestinal problems, and skin and infectious diseases give an impression of the importance of vinegar in those days. However, knowledge about the production process of vinegar was purely empirical, and it was

Ullmann's Food and Feed
© 2017 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim
ISBN: 978-3-527-33990-7 / DOI: 10.1002/14356007.a27_403.pub2

not known that the production was catalyzed by acetic acid bacteria. Today, it is known that wild-type acetic acid bacteria can be found in combination with yeasts on almost all sugar-containing fruits and in plant juices, and thus also in liquids containing alcohol. Additionally, they are transferred by insects, in particular by the fruit fly (*Drosophila melanogaster*). Even without knowledge of the microbiological nature, production and trade could be well developed over the centuries. Vinegar production in Europe first took on pre-industrial shape in France, near the end of the 1300s. The technical evolution of production systems based on the increasing understanding of the phenomenology and later scientific discoveries enabled further industrialization of vinegar production.

The scientific knowledge of the causal relations of the conversion from ethanol to acetic acid is still relatively young and was collected over a period of roughly four centuries. The significant role of air was first described by Becher (1635–1682). After the discovery of the oxygen in the air in 1773, the influence of atmospheric oxygen on the formation of vinegar was detailed by Lavoisier in 1793. Kützing (1807–1893) examined the vinegar process microscopically and found in 1838 that the acidification is caused by microorganisms ("*Ulvina aceti*") [1]. This theory had many opponents, the most famous of whom today is probably Liebig (1803–1873).

In 1864, Pasteur (1822–1895) reported experiments which revealed the influence of the microorganisms. He heated alcoholic mashes, which then failed to develop acetic fermentation, and then inoculated them with the solid fraction of an operating fermentation process, the so-called mother of vinegar ("*Mycoderma aceti*"), which resulted in acidification. The studies on vinegar in Études sur le Vinaigre published by Pasteur [2] are now considered to be the first definitive work on the microbiology of vinegar fermentation.

3. General Aspects of Acetic Acid Fermentation

The key process in vinegar production is the biocatalytic conversion of ethanol to acetic acid in aqueous solution. The incomplete oxidation is catalyzed by living cells of extremophilic acetic acid bacteria, which can survive at very low pH values in the fermentation medium. The conversion is strongly dependent on the availability of oxygen and must be carried out at moderate temperatures.

3.1. Stoichiometry

Biological vinegar production from sugar-containing agricultural raw materials is achieved by a two-step process. The conversion of sugar to ethanol and the subsequent production of acetic acid from ethanol are both fermentation processes. Ethanol is produced with the aid of yeast cells in anaerobic alcohol fermentation.

$$C_6H_{12}O_6 \xrightarrow{\text{Yeasts}} 2\,C_2H_5OH + 2\,CO_2 \quad \Delta H = -230\ \text{kJ}$$

In the next production step, the vinegar fermentation, acetic acid is produced through incomplete oxidation of ethanol to acetic acid under aerobic conditions.

$$C_2H_5OH + O_2 \xrightarrow{\text{Acetic acid bacteria}} CH_3COOH + H_2O \quad \Delta H = -493\ \text{kJ}$$

The stoichiometry leads to the following mass equation:

$$46\ \text{g ethanol} + 32\ \text{g oxygen} \rightarrow 60\ \text{g acetic acid} + 18\ \text{g water}$$

Consequently, 60 g of acetic acid is produced from 46 g of ethanol. The density of ethanol at 20°C is 0.78924 g/mL, so 58.28 mL of ethanol gives 60 g of acid (100 L of ethanol gives 102.95 kg of acetic acid). With the concentration units typically used in the vinegar industry for ethanol (vol% = mL/100 mL) and acetic acid (wt% = g/100 mL), the theoretical conversion can be determined by using the factor 1.0295. In the vinegar industry a unique definition of a "total concentration" as the sum of acetic acid concentration and ethanol concentration is used. Even though this addition of concentrations with different units (% w/v and % v/v) is incorrect from the physical point of view, the total concentration is an important variable for industrial practice.

3.2. Microbiology

Acetic acid production from ethanol is biologically catalyzed by Gram-negative and strictly aerobic bacteria of the *Acetobacteraceae* family and the genera *Acetobacter* or *Gluconacetobacter* and *Gluconobacter* with numerous individual strains [3]. The strains currently used for industrial purposes in submerged fermentation have only little in common regarding their phenomenology with wild-type acetic acid bacteria or the early production organisms. Industrial cultures can only grow in the presence of comparatively high concentrations of ethanol and acetic acid and are very sensitive to changes in process conditions. The morphology of industrially used acetic acid bacteria is rod-shaped with a length of 0.9–3 μm and a diameter of 0.5–0.8 μm. If the bacteria are cultivated at lower acetic acid concentrations, bloated and sickle-shaped cells are also found. It is assumed that much of the energy available in the cell is used to remove the undissociated acetic acid from the cytoplasm [4]. An energy deficit caused by lack of oxygen leads to uncontrolled influx of acetic acid and damage to the cell. Therefore, sufficient oxygen supply to the bacteria is essential for their viability at higher acidities. The temperatures at which acetic acid bacteria are cultivated in industrial vinegar production are typically in the range of 28–35°C. For use in tropical countries and for optimizing cooling efficiency a number of efforts have been made to shift the temperature range to higher temperatures [5].

Identification of Strains. In the vinegar industry acetic acid bacteria are not used as defined pure cultures but as a more or less diversified mixed culture. Nevertheless, especially in raw materials with low complexity like industrial alcohol vinegar, one predominant strain will be selected during long-term fermentation. Strains isolated from European industrial facilities with different processes were mainly identified as belonging to the genera *Acetobacter* and *Gluconacetobacter*, e.g., *Acetobacter aceti*, *Acetobacter pasteurianus*, *Gluconacetobacter hansenii*, *Gluconacetobacter xylinus*, and *Gluconacetobacter europaeus* [6, 7]. The identification of a phylogenetic tree of acetic acid bacteria by sequence analysis of amplified 16S rRNA has proven to be a reliable technique to identify homologies of the individual strains [8]. Interestingly, also in traditional Japanese fermentation a monoculture of *Acetobacter pasteurianus* could be found by using this type of method, whereas a more complex biocoenosis was expected [9]. If it is not certain whether a desired culture has persisted in the fermenter after inoculation, a simple plasmid sample analysis can be employed to identify the culture.

Genetic Stability of Acetic Acid Bacteria. Industrial acetic acid bacteria can change their characteristics and even lose their capability for ethanol oxidation when cultivated in other cultivation systems without sufficient selection pressure. This can happen during microbiological operations on agar plates or in shaking flasks, in modified media, or even in standardized fermentations with merely different concentration ranges. This tendency to revert under modified selection pressure can be explained by the extraordinarily high genetic instability of acetic acid bacteria. Insertion sequences (ISs), that is, transposons or jumping genes, are partly responsible for this. The first studies on the 1380 IS element, which can jump to the position of the gene for the alcohol dehydrogenase and turn it off, have in the meantime been extrapolated to other IS elements and confirmed [10].

It is possible to select mutants and production cultures with enhanced acid resistance and product formation rates in laboratory-scale fermenters by using their own genetic variability and methods of directional selection. For this reason and because of several restrictions in the food industry, no genetically modified microorganisms are used today.

Key Enzymes and Energy Metabolism. The fast dehydrogenases located on the outside of the cytoplasm membrane are responsible for the extraordinarily high specific ethanol conversion rates of the acetic acid bacteria. They contain pyrroloquinoline quinone (PQQ) as prosthetic groups [11]. The membrane-bound alcohol dehydrogenase (ADH) and acetaldehyde dehydrogenase (ALDH) are the dominant catalysts for the oxidation of their substrates in the periplasm [12]. The cytoplasmic NAD-dependent dehydrogenases have significantly lower enzymatic activity. The positions of the dehydrogenases and the respiratory chain are

predestined for the fast oxidation of large quantities of the substrate. This is necessary to maintain the energy metabolism, since the energy gain from the incomplete oxidation of ethanol is low. Acetic acid bacteria metabolize sugars such as glucose for use as carbon source via the hexose monophosphate pathway (HMP) and the citric acid circuit [13]. However, some acetic acid bacteria can also oxidize glucose to gluconate and 2-ketogluconate at the membrane-bound enzymes in the absence of ethanol [3].

Theories on Acid Resistance. Acetic acid bacteria are extremophiles that are viable under very harsh conditions. Like thermophiles or halophiles these acidophilic microorganisms have evolved certain strategies and protective mechanisms to survive under the highly noxious conditions in the culture. However, the acid tolerance of acetic acid bacteria is not fully explained. Different approaches and theories are based on chemical groups that limit the intrusion of acid into the cytoplasm. Hopanoids, which are similar to sterols, can be found with various substituents [14] in significant concentrations, predominantly in cells that were exposed to high acid stress under industrial conditions. Over-expression by using a multicopy plasmid resulted in an increased acid tolerance. Acetic acid bacteria have a specific proton gradient pump and thus an efficient transport mechanism for acetic acid as a protective mechanism [4]. Under laboratory conditions the concentration of the enzyme aconitase increases with rising acid concentration [15].

Starter Cultures and Production of Monocultures. The mash for industrial submerged-culture vinegar fermentation is inoculated with liquid starter cultures. Typically, these starter cultures are taken from fermenters running with the same raw material and process modes. For high-acidity fermentations it is not possible to cultivate acetic acid bacteria in special seed fermenters with high biomass concentrations. Owing to their genetic instability, they would start to change their properties once cultivated under different process conditions. Therefore, the biomass of an active fermentation process is sampled at an exactly defined point in time when the bacteria are in exponential growth. The samples are then blended with a protective nutrient combination and, if taken at high total concentration, diluted. Such cultures can be stored over several months in their persistent form at room temperature. The inoculation of an industrial fermentation system is then carried out on the basis of a starter recipe with a comparatively small starter culture volume of 5–10 L for 100 m^3 working volume without preculture stages.

Recently, lyophilization methods for the long-term preservation of acetic acid bacteria starter cultures have been the subject of intense research, and the enzyme activity of lyophilized acetic acid bacteria has been analyzed [16]. Novel starter cultures for industrial purposes have been prepared by neutralization and cryopreservation with special cryoprotectors have also been investigated [17]. These techniques have not yet applied in the vinegar industry due to technical problems, in particular with the preservation of high-acidity cultures, and poor economic feasibility. Not only is it difficult to secure mixed cultures, but also to separate individual monocultures from mixed cultures microbiologically. However, special dual-layer agar plates [18] with customized media compositions can be used to handle low-acidity strains.

Contamination. The acidity of vinegar and the process conditions effectively prevent typical food contaminants growing and damaging the raw product during fermentation. In the industrial practice of alcohol vinegar production, acetic acid concentrations typically exceed 10% and can reach up to 23%. The process is inherently sterile, and this has a significant impact on the hygienic design requirements of a process plant. The production of wine vinegar or other extract-rich vinegars is more prone to contamination due to the lower acidity and the complex, nutrient-rich matrix. With suitable hygienic measures, contaminations in the processing area with yeasts, lactobacilli, fungi, etc. and the resulting losses in yield and quality can be avoided. The same applies to the occurrence of fruit flies (*Drosophila melanogaster*), vinegar eels (*Turbatric aceti*), and mites. Recontamination with vinegar bacteria is possible if filtration and pasteurization are inadequate. This can finally cause the undesired formation of the typical layer on the liquid surface of the commercial vinegar products. Contamination of vinegar with

lower acidities can also be caused by acidophilic fungi such as *Moniliella acetoabutans*, which can be prevented by filtration, pasteurization, or the addition of chemical preservatives [19].

Submerged-culture fermenters can be contaminated with bacteriophage infections [20]. By screening for bacteriophage-resistant cultures and operational measures in raw-material management, these contaminations and serious production shut downs can be avoided.

3.3. Mixing and Aeration

Oxygen plays an important role as essential cosubstrate in vinegar fermentation. Even short interruptions of oxygen supply will harm acetic acid bacteria and decrease productivity in fermentation processes [21]. Oxygen must pass the phase boundary between air and the liquid in which the acetic acid bacteria are cultivated, whether this is a traditional surface culture or a highly sophisticated submerged-fermentation process. For submerged fermentation the phase boundary is defined by the surface of the dispersed bubbles in the fermentation broth. The mass transfer of oxygen into the vinegar broth can be calculated by using the straightforward equation for the oxygen-transfer rate (OTR)

$$\mathrm{OTR} = k_L a(O^* - O)$$

$k_L a$ = specific mass-transfer coefficient
O^* = saturated concentration of dissolved oxygen (= concentration at phase-boundary surface)
O = dissolved oxygen concentration in liquid

If the oxygen uptake rate (OUR) of the acetic acid bacteria is not limited, the productivity of a vinegar production process is directly linked to the OTR, and the higher the specific mass-transfer coefficient $k_L a$ the higher the productivity. The $k_L a$ value of a vinegar broth can be enhanced by high-performance aeration systems, which increase the specific bubble surface area a by providing sufficient air flow rates with a fine bubble dispersion. Thus, the design of the mixing and aeration system plays an important role in vinegar fermentation technology. In particular, the specific local energy dissipation has a significant influence [22].

The saturation concentration of oxygen O^* at the gas/liquid phase boundary is

$$O^* = \frac{y_{O_2} \cdot p_{\mathrm{abs}}}{He}$$

y_{O_2} = mole fraction of oxygen in gas
p_{abs} = pressure in tank (head pressure + hydrostatic pressure)
He = temperature-dependent solubility constant (Henry coefficient)

For a given fermenter size and geometry, O^* could be increased by increasing the partial pressure of oxygen in the feed gas (oxygen-enriched air) and by increasing the head pressure (by use of pressure vessels). However, both options are typically not used in the international vinegar industry for economic reasons, even though academic studies have shown their potential [23]. Increased oxygen transfer requires a higher cooling capacity due to the direct correlation of heat dissipation with oxygen consumption according to the stoichiometry.

3.4. Fermentation Kinetics

The process design for vinegar production must consider a number of basic kinetic properties and restrictions. Even when the development of new production mutants and technologies allows operating at higher acetic acid concentrations and with higher productivities the following aspects are generally valid:

1. Substrate inhibition by ethanol: Ethanol is a toxic substrate for acetic acid bacteria, and the ethanol concentration must not exceed a certain threshold.
2. Substrate limitation by ethanol: Ethanol is the energy source for the bacteria, and metabolic activity cannot be maintained without a sufficient amount of ethanol in the fermentation broth.
3. Product inhibition by acetic acid: At high concentrations of acetic acid the metabolic rate decreases.

4. Product "limitation" by acetic acid: Mutants of acetic acid bacteria cannot grow at low acetic acid concentrations.
5. The "total concentration" must be kept in a certain range: Even minor transgressions of the culture specific limit can have significant consequences for process activity.
6. Oxygen limitation: Oxygen is obligatory and necessary for any metabolic activity.
7. No significant CO_2 formation: The incomplete oxidation of ethanol to acetic acid does not involve the citric acid circuit or other pathways with CO_2 formation (at higher total concentrations).
8. No growth on ethanol: Carbon from ethanol cannot be used as source for anabolic pathways and a secondary carbon source such as glucose is needed.
9. Partial/nongrowth-associated product formation: Product formation occurs during growth and also without growth of acetic acid bacteria.
10. Residual activity of nonviable cells: Even cells with limited viability can maintain the ability to catalyze acetic acid formation with their membrane enzyme system.
11. Restricted temperature range: Optimal fermentation temperatures for growth at low acidity and for resistivity against high acid concentration vary.

The above phenomenological characteristics must be taken into account for efficient process design and optimization, but also for practical industrial process operation based on "expert knowledge".

For the analysis of the dynamic behavior with mathematical equations a set of first-order differential equations can be formulated and solved numerically [24]. Reaction velocities r_i for substrates and products and r_{X_i} for the biomass

$$r_i = \left.\frac{\mathrm{d}C_i}{\mathrm{d}t}\right|_{\text{Reaction}} \qquad r_{X_i} = \left.\frac{\mathrm{d}X_i}{\mathrm{d}t}\right|_{\text{Reaction}}$$

can be expressed in mathematical formulas to assemble a kinetic model [25]. One of many possibilities for modeling acetic acid fermentation [26–31] is described below to show the general concept:

- Differentiation of volumetric concentrations of viable X_v and nonviable X_{nv} cells, both of which have catalytic activity. Only the viable cells can grow.

$$X = X_v + X_{nv}$$

- Growth rate μ_{EtOH} on ethanol (C_{EtOH} = volumetric concentration of ethanol) is negligible.

$$\mu = \mu_{Glu} + \mu_{EtOH} \quad \mu_{EtOH} \approx 0$$

- However, growth μ_{Glu} on glucose (C_{Glu} = volumetric concentration of glucose) is limited by ethanol and oxygen (C_{O_2} = volumetric concentration of oxygen) due to lack of energy. Acetic acid (C_{HAC} = volumetric concentration of acetic acid) and ethanol inhibit growth.

$$\mu_{Glu} = \mu_{Glu,max} \cdot \underbrace{\frac{C_{Glu}}{C_{Glu} + K_{m,Glu}}}_{\text{Limitation}} \cdot \underbrace{\frac{C_{O_2}}{C_{O_2} + K_{m,O_2}}}_{\text{Limitation}} \cdot \underbrace{\frac{C_{EtOH}}{C_{EtOH} + K_{EtOH}}}_{\text{Limitation}} \cdot \underbrace{\frac{K_{I,HAC}}{K_{I,HAC} + C_{HAC}}}_{\text{Inhibition}} \cdot \underbrace{\frac{K_{I,EtOH}}{K_{I,EtOH} + C_{EtOH}}}_{\text{Inhibition}} \cdot \ldots$$

where K_I is the inhibition constant and K_m the half-saturation coefficient.

- The death rate k_{dv} of viable cells is slightly accelerated by high concentrations of ethanol and acetic acid and abruptly initiated when ethanol is depleted ("zero run") and when aeration is stopped.

$$k_{dv} = k_{dv,min} \cdot \underbrace{\left(1 + e^{k_{EtOH} \cdot (C_{EtOH} - E^*)}\right)}_{\text{Lost viability at high EtOH}} \cdot \underbrace{\left(1 + e^{k_{HAC} \cdot (C_{HAC} - A^*)}\right)}_{\text{Lost viability at high acid}} \cdot \underbrace{\left(1 + \frac{E^{**}}{C_{EtOH}}\right)}_{\text{'Zero' run}} \cdot \underbrace{\left(1 + \frac{O^{**}}{C_{O_2}}\right)}_{\text{Lack of aeration}} \cdot \ldots$$

$$k_{dnv} = \gamma(C_i) \cdot k_{dv}$$

where γ is the death-rate proportional coefficient.

- The reaction velocities can be expressed as follows, where $Y_{i/j}$ are yield parameters proportional to stoichiometry.

$$r_{X_v} = \mu \cdot X_v - k_{dv} \cdot X_v$$

$$r_{X_{nv}} = k_{dv} \cdot X_v - k_{dnv} \cdot X_{nv}$$

$$r_{Glu} = -\frac{1}{Y_{X_v}} \cdot \mu \cdot X_v$$

$$r_{HAC} = \alpha \mu X_v + \beta (X_v + X_{nv})$$

$$r_{EtOH} = -\frac{1}{Y_{HAC/EtOH}} \cdot r_{HAC}$$

$$r_{O_2} = -\frac{1}{Y_{HAC/O_2}} \cdot r_{HAC}$$

where α and β are the product formation coefficients, and μ the specific growth rate.

- The kinetic model can be merged with a reactor model like the one for an ideal mixed vessel (modification for scales with large volumes V_R is necessary).

$$\frac{d(V_R C_i)}{dt} = V(C_{i0} - C_i) + r_i V_R$$

The fixed model parameters (e.g., K_i, k_i, i^*, . . .) of these models can be determined by comparison of real experiments with computer simulation and the use of algorithms to minimize the deviation between the data [32, 33]. The general problem with fixed-parameter models is the fact that acetic acid bacteria can mutate under changing process conditions. Therefore, the quantification of model parameters is only valid for a narrow range of operational process settings.

Alternative models have been developed that take into account variations of RNA content of the cells and other alternative state variables [34, 35]. Some processes for the production of high acetic acid concentrations use temperature profiles during fermentation [36]. Therefore the dependence of the model parameters on the actual temperature must be taken into account, too. However, even with these modifications, such models are suitable for general process design or training only and not for predictive simulation or model-based automation.

4. Resources of Raw Materials

According to the main reaction, ethanol and oxygen are the substrates for the production of acetic acid in an aqueous solution. Sugar, in particular glucose, is needed as cosubstrate for growth. Nutrients for the supply of nitrogen, phosphorus, potassium, and other elements for growth and metabolism of acetic acid bacteria are additionally needed. Amino acids, vitamins, and other growth factors can stimulate growth, catalytic activity, and resistance against high acidity [37].

Alcohol Vinegars. Most of the vinegars produced worldwide are alcohol vinegars (also called white vinegars or spirit vinegars) produced from highly purified alcohol of agricultural origin. The alcoholic mash as a mixture of azeotropic ethanol (96%) and process water does not contain sufficient nutrients for fermentation. Specific nutrient recipes are used in the industry to provide high fermentation performance with a minimum quantity of added nutrients. For tax reasons, in most countries the alcoholic mash must be denatured either with acetic acid/vinegar (e.g., in Europe) or ethyl acetate (e.g., in the USA).

Extract-Rich Vinegars. In the second group of raw materials for vinegar fermentation, the alcoholic mashes are derived from complex raw materials without significant purification steps like rectification. These extract-rich raw materials are derived from fruits, vegetables, roots, or grains containing polysaccharides or sugars which have been alcoholically fermented prior to acetous fermentation. Typically the basic raw material determines the character of the final product in terms of color, constituents, and taste. Most of these raw materials contain sufficient nutritional components for bacterial growth. In some cases, a limited amount of nutritional salts must be added. While fermenting the raw materials in alcoholic and acetous fermentation, some components are consumed, some are evaporated or precipitated, and several derivatives of the wide range of primary constituents are produced. The final vinegars produced from different agricultural sources can contain complex carbohydrates, higher alcohols, esters, aldehydes, ketones, organic acids, salts, amino acids,

polyphenols and other phytochemicals, and their derivatives [38].

5. Production

5.1. Processes

5.1.1. Submerged Fermentation

Submerged processes are currently the predominant type of fermentation used in the industrial production of vinegars worldwide. The basic Acetator technology dates back to an invention in the late 1940s [39] and has been engineered and optimized by the German company Frings since then. Today this technology allows fully automated operation with high productivity and conversion from ethanol to acetic acid near to the theoretical yield.

This type of fermentation is performed in large fermentation vessels filled with a liquid phase with dispersed air bubbles and with suspended cells. The rheology of the fermentation medium is characterized by water-like viscosity and the absence of a solid phase.

Typically, fermentation is operated according to the following process modes:

1. Continuous processes
2. Repeated batch processes
3. Repeated fed-batch processes
4. Dual-stage repeated fed-batch processes

In *continuous processes* the growth rate μ of the biomass must be equal to the volumetric flow rate through the fermentation vessel to avoid a "washout" of the cells. However, at high acetic acid and low ethanol concentrations, acetic acid bacteria are not able to grow at a sufficient rate. Therefore, single-stage continuous processes can only be applied at low acetic acid concentrations (< 10%).

Industrial vinegar fermentation processes are commonly carried out in processes with discontinuous fermentation cycles and with varying concentrations of cell mass, ethanol, and acetic acid over time. The cycles follow one another, with only a partial volume being discharged from the fermenter at the end of each cycle. The biomass contained in the remaining portion of the fermentation medium serves as the inoculum for the subsequent cycle after charge with new raw materials. The cycle times can vary from 10 to 70 h depending on the process mode, the ethanol concentration of the raw material, and the process parameter settings. In the *repeated batch process*, also called standard process, the fermentation volume and the total concentration are constant over the fermentation time. Acidities of up to 15.5% w/v acetic acid can be achieved. At the end of a cycle around 30% of the fermenter volume with the final acidic product and a final ethanol concentration of 0.2–0.4% is discharged. After charging with a new alcohol-containing mash with a total concentration equal to the previous batch, the new cycle starts with an ethanol concentration of 2–5% (depending on the total concentration). The exact balance for the process setup is calculated in advance, taking into account the mash concentration of the individual raw material and correction factors, e.g., for evaporation losses.

The *repeated fed-batch process* which is known in the industry as the *single-stage high-strength process* (SSHS) is a modification of the standard process. The process starts after charging with a moderate total concentration and a lower ethanol concentration in comparison to the standard process. Therefore, conditions are adjusted so that the acetic acid bacteria can find ideal conditions for recovery from the last batch and for optimal growth. During the process concentrated ethanol is added to increase the total concentration to a precalculated value. In the last stage of the fermentation, the remaining ethanol is converted to acetic acid, and the discharge pump is activated when the ethanol content reaches the final concentration. The discharged volume can exceed 70% of the final volume, depending on the final acidity and the concentration of the ethanol mash for the intermediate feeding. The SSHS process is used for alcohol-vinegar production only. The final acidity can reach up to 23% acetic acid. The mean specific productivity in sophisticated fermentation apparatuses like the Frings Acetator amounts to 40–60 $kg_{acetic\ acid}\ m^{-3}\ d^{-1}$. An alternative process design with the aim of decoupling growth and final conversion of acid independent from further viability of the culture was developed [40, 41]. High acidity levels with a two-stage fermenter cascade operating in a repeated fed-batch system

(dual-stage high-strength process, DSHS) have been observed. However, due to the lack of genetic feedback from the second stage into the first stage the process was not as easy to use for in-process screening purposes as the SSHS process. Today, the SSHS process can produce higher acidities and is used preferentially also because of its simplicity. The maximum specific growth rate of acetic acid bacteria in alcohol-vinegar fermentation with high acidities (> 17%) typically does not exceed values of $\mu = 0.05–0.07\ h^{-1}$. These growth rates are much lower than in fermentations with reduced acidities or total concentrations. The latter can reach > $0.4\ h^{-1}$ when the total concentration is less than 5% [21]. Growth rates are not only dependent on total concentration but also on the raw material and the available nutrients therein. Growth in SSHS processes is characterized by a short logarithmic growth phase followed by a longer linear growth phase and a brief stationary phase. In high-acidity fermentations, a pronounced death phase with decreasing biomass concentrations occurs with increasing acetic acid concentration. This is the case in particular with SSHS processes for producing alcohol vinegars with acidities close to the acetic acid tolerance of the bacteria. The maximum biomass concentration in the process is rather low. In alcohol-vinegar fermentation the biomass concentration reaches only 80–150 $mg_{dry\ biomass}/L$. In extract-rich vinegars this concentration is higher but does typically not exceed 1 $g_{dry\ biomass}/L$. Compared to other industrially used microorganisms acetic acid bacteria are characterized by an extraordinarily high specific product formation rate. This rate can reach up to 25 $g_{acetic\ acid}\ g^{-1}_{dry\ biomass}\ h^{-1}$ in industrial systems during the logarithmic growth phase. Even during the death phase with high acidities, values $> 15\ g_{acetic\ acid}\ g^{-1}_{dry\ biomass}\ h^{-1}$ are achieved.

The semicontinuous cyclic operation can be carried out over a period of several months without interruption. As long as the changes in high and low acetic acid and ethanol content are reproducibly adjusted within each cycle and in a tolerable concentration window the biomass will not change its characteristics. This specific kind of operation allows in-process selection of bacteria which maintain their acid resistance and are able to grow. Mutants with lower performance are "washed out" over time. Consequently, in particular for SSHS and DSHS acetic acid fermentations, it is absolutely necessary not to change these conditions to lower acidities for a longer period of time. Otherwise different strains or mutants, adapted to the new process conditions, will overgrow the former predominant population.

5.1.2. Surface Fermentation and Trickle Processes

Today, surface fermentation processes are used only for highest quality culinary vinegars. In principle these technologies are identical to or derived from ancient surface processes. In these the fermentation liquid is covered with so-called mother of vinegar, a layer at the surface of the liquid consisting of a cellulosic structure formed by bacteria and the bacteria themselves. Oxygen supply from air is restricted to the surface, and therefore the productivity is very limited. This simple technology was used in early times in single vats and later in batteries of wooden or sandstone vessels. The Orleans process was one of the first industrialized production methods. In the 1600s, towerlike production vessels were developed in which the bacteria were immobilized on wood chips with a higher specific surface area. For the Boerhaave process, the liquid to be fermented was manually poured, and, in later developments in the 1800s like the Schutzenbach process, trickled over the filling. An excellent summary of all aspects of these historical vinegar production processes can be found in [1]. The further development that determined the state of the art until the mid-1900s was the so-called Frings Generator. This was the last real innovation of the trickle fermenters working with wooden chips before submerged fermentation was invented. However, several of these old devices are still in operation after more than 40 years. Also surface fermentations are still in use for high-quality vinegars in Japan, China, and Europe. In particular the well-known high-price vinegars Aceto Balsamico di Modena traditionale and the Vinagre de Jerez are produced with the aid of multistage batteries of small wooden vessels. However, the volumetric amount produced by traditional methods is negligible in comparison to submersible fermentation.

5.1.3. Solid-State Fermentation

Solid-state fermentation (SSF) was the predominant fermentation type in China, and still plays an important role in production processes. This old technology has not been frequently used in Europe or other countries apart from Asia. Vinegars produced by SSF and aged with different methods are known as traditional vinegars rich in typical smell and flavor. A number of different raw materials such as rice, corn, wheat, barley, and broomcorn are fermented by SSF. As opposed to the western world, where the Frings Acetator process represents a kind of "standard" and is the most common production method, there is a huge variety of industrial SSF processes for different vinegars [42]. An example with glutinous rice as main raw material is described below.

In order to produce typical vinegar from glutinous rice additionally wheat bran, rice husks, salt, sugar, yeast, enzymes, and a wheat starter culture (mixture of bacteria and different types of molds like *Aspergillus spec.*, *Mucor spec.*, and *Rhizopus spec.*) are needed. First, rice is soaked in water before it is steamed and further processed. After addition of water, enzymes, yeasts, and the wheat starter, saccharification and alcoholic fermentation of the rice occurs in closed fermentation vessels. This process can take more than 10 d. The alcoholic solution is then pumped into large open vats, where it is mixed with a sufficient amount of vinegar starter (from the last batch) and wheat bran, and covered with rice husks to give the solid-state bed. During 12–35 d of acidification the bed is mixed several times and water and rice husks are added. The raw vinegar which drains to the bottom of the vats is sampled and trickled to the solid bed, too. Mixing and water addition are carried out to provide a fermentation temperature of 35–45°C and a moisture content of about 60% and to provide sufficient access of atmospheric oxygen to the bed. The medium is anaerobically matured for several days and then filtered through bamboo mats to obtain liquid vinegar. Sugar is added and the mixture stored for several months for maturation and improvement of taste and color. The final product is filtered again and then pasteurized before bottling. The products are typically of dark color and contain a huge variety of components extracted from rice and produced during fermentation.

5.1.4. Alternative Technologies and Trends

Several attempts in industry and academia have been initiated in the past to further optimize the processes for vinegar fermentation. Reviews of these approaches until the 1990s can be found in [43, 44].

Surface fermentation is used in industry for high-price vinegars only. The original old production method and the long-term maturation is part of the "marketing concept" for these flavor-rich vinegars. Some devices have been invented to provide a controlled air atmosphere [45]. From a theoretical viewpoint, new developments in surface fermentation could increase the surface area by means of bubble-free aeration with membranes inside the liquid or moving surface films. The main advantage would be that the evaporation of volatile product constituents would be minimal, but the productivity would be higher in comparison to traditional systems. However, neither the technical nor the commercial feasibility has been confirmed yet. It is doubtful that modern surface fermentation techniques will find their way into the vinegar industry. Interestingly these technologies, especially automated cellulose production with the aid of acetic acid bacteria for pharmaceutical purposes, have found their way into modern biotechnology [46].

Most of the works on traditional solid-state fermentation processes are done by manual operation or with simple mechanical devices. Automation of SSF units has been discussed, and a number of apparatuses have been proposed in the industry. However, specific empirical training and know how are necessary to produce the desired quality of the traditional vinegars on an industrial scale. It is rather difficult to reproduce the complex manual processes with their inhomogeneities in an automated technical device and achieve a product of comparable quality.

Process engineers have several opportunities to improve submerged or related bioprocesses with regard to productivity. Some examples are the recovery of biomass to increase the concentration of the biocatalyst and make the hydraulic load independent from microbial growth [47]. Regardless of the chosen method of process intensification, an equivalent increase of oxygen transfer rate (OTR) is necessary. A process using a repeated fed-batch culture in a laboratory jar fermenter with a hollow-fiber membrane filter for

cell recycling has been studied [31]. A mathematical model (see Section 3.4) balancing viable and nonviable cells has been formulated which fits well with the experimental data. Attempts to optimize continuous operation were also reported [48] by using columns packed with ceramic supports. A continuous pulsed-flow process in a ceramic fixed bed has been developed [49]. In some of these laboratory-scale processes remarkably high productivities of more than ten times those of industrial plants could be observed. However, on the fermenter scale, the complexity of the apparatus and the conditions with nutrient-rich media, low acidities between 2.5 and 7%, and the high aeration rates are far from industrial reality. None of these processes or the devices in which they have been carried out found their way into industrial production plants.

Aggregating and nonaggregating bacterial cultures were immobilized by transition-metal complexation in a continuously operated tower reactor [50]. The use of aggregating cultures has shown an increase in productivity. Siran, wood chips, and polyurethane foam have been used as supports for the growth of bacterial cultures to produce high quality Jerez-Xèréz Sherry vinegar [51]. Polyurethane foam with an average pore size of 400 μm has shown to be the most effective material. Other immobilization studies are reported in [52–54].

Bacterial cultures for high-acidity fermentation cannot be selected with classical microbiological methods (see Section 3.2). An intelligent approach to process optimization is the use of cognitive algorithms, e.g., fuzzy logic [55] or neural networks. A computer program estimates the status of the biomass with regard to viability and stress tolerance during the ongoing process. The current status is calculated from measurable data of previous batches on the basis of fuzzy sets. These fuzzy sets and the interconnection between them reflect, translated into mathematical equations, the scientific knowledge, and empirical expert knowledge about the process. From this information a new recipe for the next batch is calculated and processed by the central PLC automation system.

An interesting approach to get more information from laboratory trials and for further process optimization is the use of parallel fermentation units. Small-scale cultivation with a new shaken-flask technology is described in [56].

5.2. Facilities, Apparatus, and Key Equipment for Submerged Fermentation

The equipment needed and the design of a plant for submerged vinegar production depends not only on the vinegar to be produced but also on the integration into food production processes. For alcohol-vinegar production, three main cases can be differentiated. One is vinegar production only for bulk and bottle filling and distribution. Ethanol can be procured on the business-to-business market or from local distilleries. Forward integration can be observed in a number of distilleries in which alcohol vinegar is produced from re-diluted rectified ethanol. Vinegar production sections with backward integration are often found in the production fine food and sauces.

For wine- or fruit-vinegar production, most producers buy grape or fruit wines on the regional, national, or international market and produce vinegars with subsequent bottling or filling in other containers. Some produces the wines from juices in the plant. However, several plants exist as part of vineyards or agricultural facilities with a full integration of the value chain.

5.2.1. General Plant Design

A plant for alcohol-vinegar production (Fig. 1) requires the following sections:

- Production building
- Laboratory
- Energy supply and emergency power plant
- Cooling system
- Wastewater neutralization
- Process water preparation
- Ethanol and nutrient preparation
- Fermentation section
- Exhaust gas recovery
- Filtration unit
- Pasteurization
- Tank farm
- Storage and filling section

The investment costs of vinegar plants are relatively low compared to other fermentation facilities, e.g., for yeasts, citric acid, and amino acids. In contrast to these and many other food

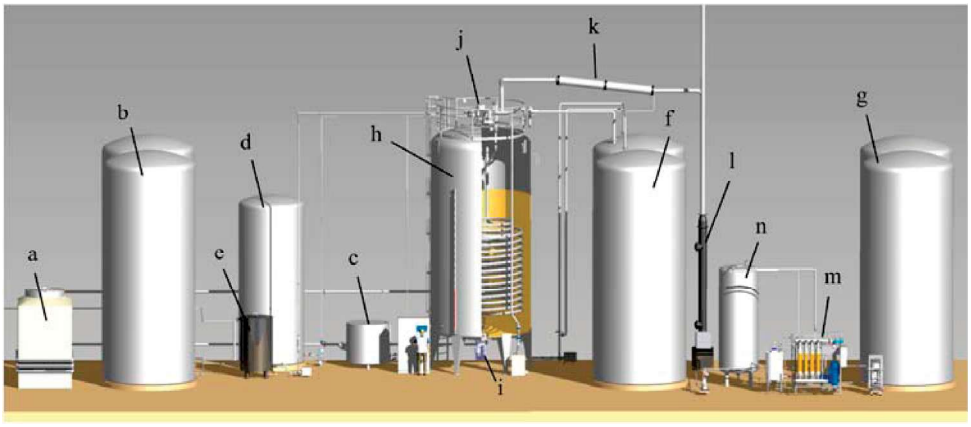

Figure 1. Schematic layout of a modern vinegar production unit
a) Cooling tower; b) Mash tank; c) Nutrient dosing system; d) Alcohol tank; e) Process water system; f) Raw-vinegar tank; g) Filtrate tank; h) Fermenter; i) Aerator; j) Defoamer; k) Condenser; l) Exhaust-gas scrubber; m) Microfiltration unit; n) Circuit tank

production processes there are no enhanced sterility requirements. Because the raw material contains ethanol and the product has a low pH value, the process is partly autosterile. The plants can be designed without sterilization-in-place (SIP) or complex cleaning-in-place (CIP) systems [57]. Other hygienic design criteria are not applied in most vinegar plants worldwide except in those which also use process steps that are sensitive to infections. Depending on local regulations the vinegar plants must use methods to prevent hygienic and other risks, such as hazard analysis and critical control points (HACCP). Nevertheless, whether a factory implements a higher degree of modern equipment with hygienic design and automation depends on the general policy of the company operating the vinegar plant. With 25–50% of the total investment costs, the tank farm for raw material and product storage and for blending plays an important role in terms of capital costs. The fermenter section with 20–30% and the filtration section with 4–6% can be calculated roughly to cost one-third of the total investment.

5.2.2. Fermentation Systems

The most important technology for industrial submerged-fermentation vinegar production is the Acetator system [58]. Worldwide, more than 60% of vinegars are produced in these fermentation systems or in generic copies based on this technology. These fermenters consist of a cylindrical vessel equipped with:

- Rotor/stator turbine for mixing and aeration
- Heat exchanger for temperature control
- Foam centrifuge for liquid recovery from exhaust gases
- Air inlet filter
- Exhaust-gas condenser
- Exhaust-gas scrubber
- Pumps and piping
- Sensors and actuators

The maximum working volume of the industrial-scale fermenters reaches up to 100 m^3. Larger systems can be engineered and manufactured, but the demand in industry is more related to modular use of fermenters. This allows greater flexibility in terms of fluctuations in required production capacity and the use of different raw materials. Small fermentation units with < 20 m^3 are used in small companies, but also in larger companies to produce specialty vinegars. However, the larger the fermenter, the lower the product specific investment and production costs.

Aeration Technology. The Acetator is equipped with a so-called Friborator turbine (Fig. 2) [59]. It operates as a mixing and aeration device, and due to the self-aspiration no

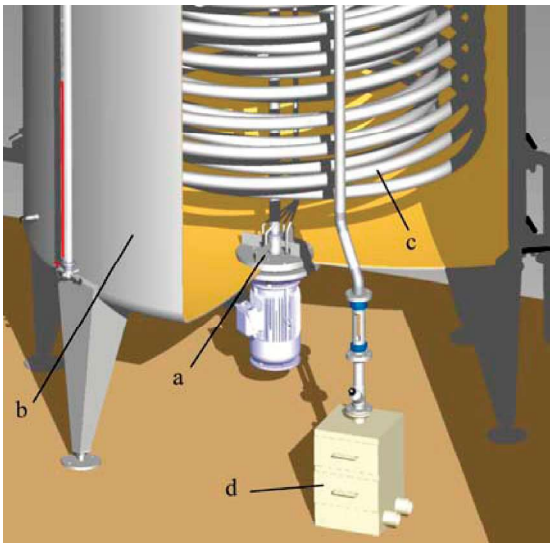

Figure 2. Installation of a self-aspirating aeration turbine (a) in a Frings Acetator tank (b) with internal cooling coils (c) and air inlet filter (d)

additional compressor is needed to provide an air flow into the fermentation vessel. The turbine consists mainly of a hollow rotor surrounded by a stator. The star-shaped rotor is installed with a direct motor drive at the bottom of the fermenter tank. The rotor draws in liquid axially and accelerates it onto the working surfaces radially and tangentially. During rotation, an area of partial vacuum is created on the rear side of the working surfaces, which are connected to a central air pipe. Because of the difference in pressure in relation to the atmospheric air in the pipe, gas flows out into the rotor's working area and is mixed intensively with the liquid. The liquid–air mixture is conveyed to the channels of the stator. The mixture flows out of the channels as turbulent jets (Fig. 3).

The length and geometry of the channels depend on the tank geometry and the required performance data. The rotor turns with a nominal rotation frequency of 1480 or 1780 rpm. High local energy dissipation creates extremely fine primary bubbles with a high specific surface area a in the aerator. The medium, containing acetic acid and ethanol, is fully coalescence inhibited, that is, small gas bubbles do not form bigger bubbles in the broth while they rise to the surface and provide excellent mass transfer of oxygen into the liquid. A high specific mass transfer coefficient $k_L a$ at a given air flow rate and therefore a high integral oxygen transfer rate (OTR) can be achieved. The theoretical achievable productivity of the fermenter is directly linked to the OTR.

Acetic acid bacteria cultures are obligate aerobic and at high acidities very sensitive to lack of oxygen. An emergency power unit, at least for the

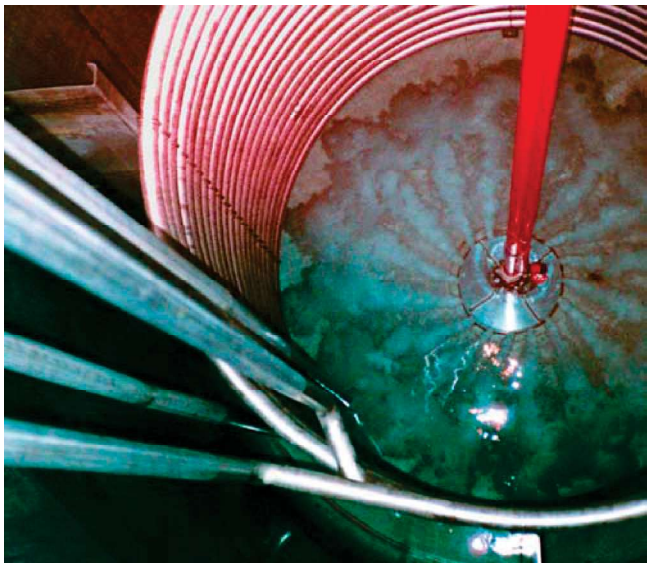

Figure 3. View into a Frings Acetator (after starting the rotor/stator turbine at reduced filling level)

motor of the aeration turbine, is essential for factories in countries where the power supply can be interrupted. Even short stops of the aeration turbine of 10–45 s can cause a decrease in productivity of the fermentation unit of up to 80% for a longer period of time [21]. However, not only the OTR determines the performance of the process with a specific bacterial culture but also the homogeneous distribution of the bubbles in the tank. The dissolved oxygen concentration O in sectors without sufficient air supply decreases and the bacteria can be damaged in correlation to their residence times in these zones. The overall productivity then is negatively influenced by these inhomogeneities. The objective was to design fermenter systems with highly optimized flow regimes. Today this can be achieved by using computational fluid dynamics (CFD, Fig. 4).

Significant changes in aerated fermentation volume during charge and discharge between individual cycles make it necessary to control the flow regime in the fermenter. By using flow control in the air pipe or a frequency converter for the turbine motor, sophisticated adjustment for each level can be achieved. The air flow rate should not exceed 0.1–0.15 vvm (volume of air per volume of liquid per minute) to avoid foam problems and to reduce evaporation of ethanol and acetic acid. The oxygen utilization of the aspirated and dispersed air in optimized fermenters under these conditions is between 60 and 90%.

Heat Exchanger and Temperature Control. Acetic acid fermentation produces metabolic heat, which must be dissipated by an adequate heat-exchange system. Typically, in high-performance submerged-fermentation systems internal heat exchangers are installed. They are designed as spiral coils to avoid disturbance of the main flow regime in the fermenter.

Foam Management and Exhaust-Gas Treatment. The aerated fermentation medium tends to foam depending on the raw materials and the physiological status of the bacteria. To avoid losses of product the foam must be treated. This is carried out without addition of chemicals, because the use of antifoaming agents has negative impacts on oxygen transfer and therefore productivity [37]. Most Acetators are equipped with Foamex foam centrifuges [60], which separate the foam by centrifugal forces. The foam

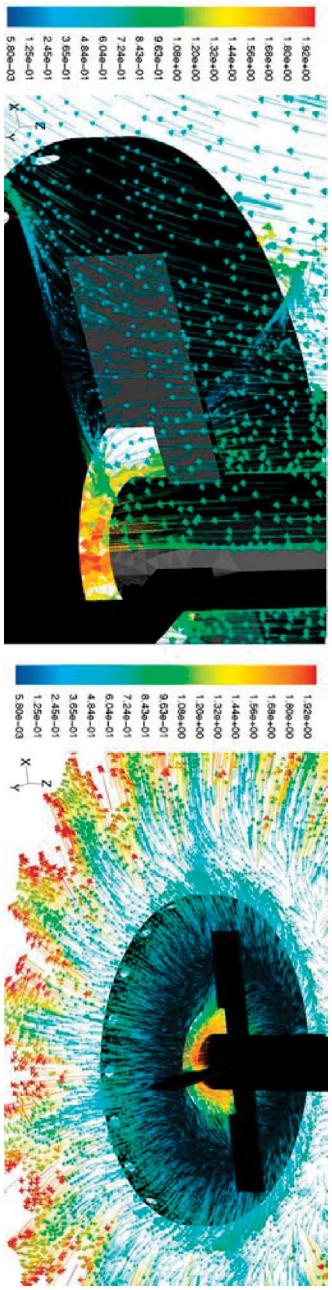

Figure 4. Typical flow regime (arrows indicate velocity vectors) in the area of an aeration turbine (computation by CFD)